

- **1. Problem Statement:**

Modern HR departments face immense challenges filtering thousands of resumes manually. This project develops an end-to-end Machine Learning system to predict whether a candidate will look for a job change, helping companies optimize recruitment workflows and reduce hiring costs.

- **2. Dataset Description:**

The model is trained on the "HR Analytics: Job Change of Data Scientists" dataset, consisting of over 19,000 candidate profiles. Features include education level, experience, company type, and training hours.

- **3. Methodology:**

- **Data Cleaning:** Imputed missing categorical values with the mode and numerical variables with the median. Unique identifier features (enrollee_id) were dropped to prevent overfitting.
- **Feature Engineering:** Converted categorical data to numerical features using One-Hot Encoding. Data was split into 80% training and 20% testing sets.
- **Modeling:** Implemented and benchmarked two classification models: Logistic Regression and Random Forest Classifier.

- **4. Results & Performance Comparison:**

The Random Forest Classifier outperformed the baseline Logistic Regression model. 78%

- **5. Advanced Business Insights & Bonus Features:**

- **Feature Importance:** Explored that training hours and city development index are the most critical factors driving recruitment.
- **Candidate Ranking System (Bonus):** Implemented a probability-based model using `predict_proba` to rank and recommend the **Top 10 Applicants** automatically.

- **6. Conclusion:**

The system effectively transitions human resources into automated AI-driven operations, saving thousands of hours in screening processes.

